Information-Parity Testing for Code Representation Effects on Frontier LLMs: Methodology and Diagnostic Findings from a Bug-Detection Pilot

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Abstract

Frontier LLMs are increasingly applied to code reasoning, yet whether the *format* of input code affects reasoning quality, independent of the *information content* of the representation, is rarely tested with controls. We present a methodology for such testing—a 5-condition design with an information-parity control (C1+) projecting a contract-augmented IR back into Python source—and instantiate it on T2 bug detection across 30 retained non-recursive Python functions, two confirmatory frontier LLMs, and a successor pair (post-confirmatory exploratory follow-on under a relaxed sampling regime, not a cross-generation replication). The primary endpoint H1 (C4 > C1+) yields a moderate positive effect (rank-biserial $r_{\text{rb}} = +0.581$, 95% CI $[+0.077, +0.917]$ excludes zero) but does not reach Holm-corrected significance under the pre-registered 9-test family (corrected $p = 0.205$). Exploratory analyses surface a directional pattern: H3 (in-text information enrichment helps) inverts on all four model-runs spanning two LLM generations. Post-hoc inspection reveals that the integer T2 composite score (range 0–3) saturates at 2—its ceiling of 3 is never attained—compressing the effective Wilcoxon n from 30 to 13–14 and constituting the structural source of confirmatory underpowering. We discuss implications for both the empirical claim and the methodology, and propose rubric refinement and T1/T3 collection as the next steps.

Contents

1 Introduction

1.1 The question

Frontier large language models are increasingly ubiquitous in developer workflows for code reasoning. Almost without exception, the input pipelines that feed them serialize code as plain source text, sometimes with light annotation. Whether plain text is the appropriate representation—and whether the *format* of the representation affects model reasoning *independently* of the information content it carries—is rarely tested with controls.

1.2 Why this is hard to test cleanly

The naive comparison—text against AST, say—confounds two variables. Two representations almost never carry exactly the same information; differences in annotations, derived edges, or type detail accompany differences in format, and any observed effect cannot be partitioned between the two axes. Prior work has either trained models on different representations (§2.1), confounding format with training signal, or applied structured prompting heuristically without controls (§2.3). To our knowledge no prior controlled study of code-representation effects on frontier LLMs combines a frozen-evaluation target, an information-parity control, and pre-registered family-wise correction. The remainder of this paper addresses that gap.

1.3 Contributions

This paper makes three contributions.

1. **A methodology for information-parity testing of code-representation effects on frozen frontier LLMs.** We define a five-condition design (§3.2.1) in which the central comparison is C4 against C1+: C1+ projects C4's typed-and-contract-annotated intermediate representation back into Python source, holding information content constant against C4 and varying only format. The parity control (§3.3) isolates the structure axis from the information axis.

2. **A pre-registered confirmatory family with formal multiplicity control.** Three hypotheses crossed with three primary metrics constitute a nine-test family, corrected by Holm–Bonferroni at $\alpha = 0.05$ (§3.6). The statistical machinery was frozen before data collection.

3. **Diagnostic findings from a 30-function instantiation.** On a T2 bug-detection task we report a primary endpoint that is direction-consistent but Holm-non-significant (§4.1); an exploratory directional inversion of H3 replicated across all four model-runs (§4.4); and a structural source of underpowering—a near-saturated location sub-score that compresses the effective Wilcoxon sample size—that motivates rubric refinement before further empirical claims (§5.3).

1.4 The single-sentence proposition

> We propose a controlled experimental methodology for measuring code-representation effects on frozen frontier LLMs, instantiate it on bug detection across 30 retained Python functions and two LLM generations, and report a primary endpoint that is direction-consistent but underpowered, alongside an exploratory directional inversion of the in-text-information-enrichment hypothesis and a structural source of underpowering inherent to the as-implemented scoring rubric.

1.5 Headline empirical findings (preview)

The primary endpoint on H1/T2 yields a rank-biserial $r = +0.581$ with 95% bootstrap CI $[+0.077, +0.917]$ and one-sided raw $p = 0.023$; under Holm correction at $m = 9$ the corrected $p = 0.205$, and the family-level decision is fail-to-reject (§4.1). The H1 direction is positive in all four model-runs across the confirmatory and successor pairs (§4.3). The H3 direction is negative in all four model-runs—opposite the pre-registered prediction (§4.4). No T2 record attains the composite ceiling of 3 (0/300 records; §3.7); the effective Wilcoxon n collapses to 13–14 on the central confirmatory cells.

Document scope. This paper is the extended preprint/journal-version draft, anticipated as an arXiv release under Constitution §14 (Phase 3 publishable outputs); a workshop-format abridgement, if pursued, would be prepared as a separate document and is not part of the present submission.

2 Related Work

2.1 Code representation for LLMs (training-time and inference-time)

Two lines of prior work bear on Lumen's question.

Training-time integration of structure. Encoder and decoder models that bake code structure into their weights through specialized training objectives—CodeBERT (Feng et al., 2020), GraphCodeBERT (Guo et al., 2021), TreeBERT (Jiang et al., 2021), and InCoder (Fried et al., 2023)—learn from ASTs, data-flow graphs, or tree-positional encodings. Lumen does not retrain; it varies the input representation supplied to a frozen frontier LLM and treats representation format as a controlled independent variable.

Inference-time structured prompting. Chain-of-thought (Wei et al., 2022), structured-output schemas, and code-sketch scaffolds introduce structure heuristically at the prompt boundary. They adopt structure as a design choice; representation format is not varied as an independent variable under explicit controls.

2.2 The closest ancestor: Namavar et al. (2022)

The direct intellectual ancestor of this work is Namavar et al. (2022), who run a controlled comparison of code representations—token sequences, ASTs, and graphs—for learning-based program repair. Lumen sits in that lineage and differs along three axes.

Target. They train generative repair models from scratch on each representation; we evaluate frozen frontier LLMs and vary only the input representation. The shift reflects the 2022→2026 deployment context for LLM-based code reasoning rather than a deficiency in their design.

Information control. Their representations differ in both format and information content; Lumen adds C1+ (§3.3), an information-parity textual control that holds information content constant against C4 and varies only format. To our knowledge no prior controlled study of code-representation effects on frontier-LLM reasoning has used such a parity control.

Discipline. They do not pre-register a confirmatory family; Lumen pre-registers H1–H3 across three primary metrics and corrects the resulting nine-test family under Holm–Bonferroni at $\alpha = 0.05$ (§3.6).

2.3 LLM prompt-format sensitivity

A separate line of work documents that frontier LLMs are sensitive to surface-level prompt-format choices independent of the underlying information content (Sclar et al., 2024; Salinas and Morstatter, 2024). Lumen lifts that concern from prompt-design folklore into a domain-specific controlled study: the format axis is operationalized as a hierarchy of code representations (§3.2.1) and the information axis is held by the C1+ parity control (§3.3).

2.4 Novelty, circumscribed

The contribution claimed here is the controlled *combination*—information-parity testing of frontier-LLM code reasoning under pre-registered family-wise correction—rather than any individual ingredient. ASTs, type annotations, contract-style specifications, frozen-LLM evaluation, and pre-registration each pre-exist; the novelty rides on the combination and on the control. Long-term inspirations from content-addressed code storage and projectional editing inform the broader Lumen research program but are out of scope for the present submission.

3 Methodology

3.1 Study phases

The study was conducted over a 90-day window (Constitution §13) in three phases.

Phase 1 (days 1–25) covered pipeline construction and dataset curation. We implemented the representation transformers, the contract generator, and the automated scoring scripts, and curated the 30 retained non-recursive Python functions. This phase produced the conversion pipelines for C1 through C4 and the C4 → C1+ projection.

Phase 2 (days 26–50) was a pilot. A pilot run exercised a small subset—approximately three functions across all five conditions, all three tasks, and both models, on the order of ninety items per run—to calibrate the runner and the automated scoring scripts. Two pilot runs were executed; the second was clean. Pilot data is not pooled with confirmatory data: the confirmatory analysis derives solely from a fresh collection on the 30 retained functions (`full_t2_confirmatory_v2`). Constitution v0.2.2 §16 (Open Question 4) anticipated a Phase-2 pilot power analysis or effect-size sanity check to decide whether 30 functions suffice or expansion to 50 is needed; no such artifact was produced, and the dataset was advanced to confirmatory collection at the constitutional minimum line of $n = 30$. The score-rubric ceiling documented in §3.7 and analyzed in §5.3 was identified post hoc from the confirmatory data; the pre-confirmatory dataset-sufficiency audit that records this gap is registered in Appendix E.

Phase 3 (days 51–90) covered confirmatory data collection (T2 only), a post-confirmatory exploratory replication on a successor frontier model pair, and analysis.

One naming artifact warrants explicit note. The runner module is `src/experiment/run_pilot.py`, named during the Phase 2 pilot and reused without modification for confirmatory data collection. The name reflects implementation history, not methodological status: the confirmatory run is not a pilot.

3.2 Conditions and dataset

3.2.1 The five conditions

Table 1 summarizes the five experimental conditions. The cumulative chain C1 → C2 → C3 → C4 progressively enriches information, while the branch C1 → C1+ enriches information in textual form. The central comparison is C4 versus C1+: both carry the same type and contract annotations and differ only in format—structured IR versus annotated Python source. This pairing isolates the structure axis from the information axis. The five conditions and their two enrichment axes are shown schematically in Figure 1.

3.2.2 Dataset composition

The dataset consists of 30 retained non-recursive single Python functions, manually curated within the 90-day scope. Their domain distribution (Appendix A) is predominantly list and array manipulation (20 of 30), with smaller representation from descriptive statistics (6), string processing (2), arithmetic (1), and control-flow or parser-like logic (1). This list-operations predominance re-

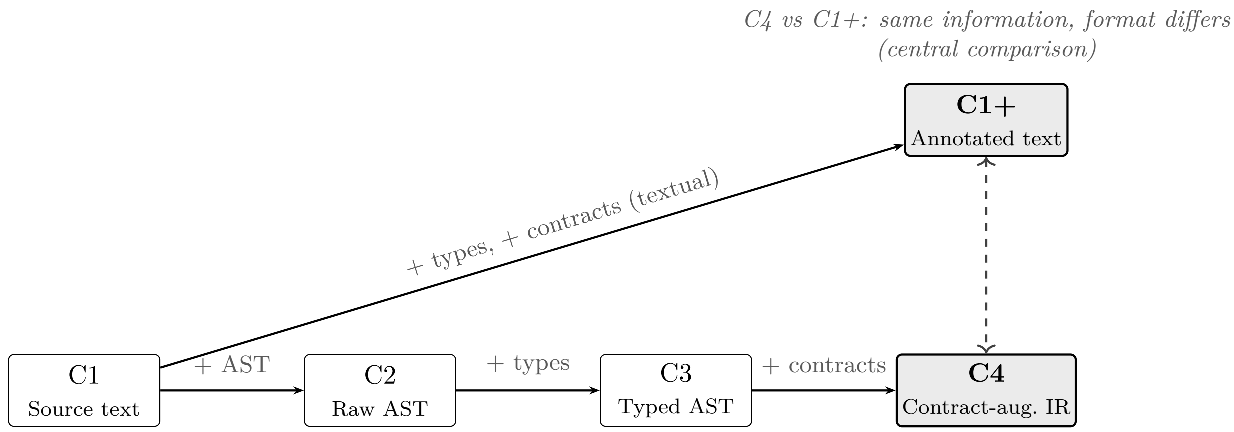

Figure 1. Schematic of the five conditions: cumulative chain C1→C2→C3→C4 and information-parity branch C1→C1+. (Optional supplementary figure.)

Table 1. The five experimental conditions. Information content is held constant between C1+ and C4 by construction; format differs (textual vs structured).

Cond	Label	Format	Info	Purpose
C1	Source text	Text	Base	Baseline
C1+	**Annotated text**	**Text**	**Enriched**	**Information-parity control**
C2	Raw AST	Structured	Base	Pure structure
C3	Typed AST	Structured	+ Types	Type effect
C4	Contract-augmented IR	Structured	+ Contracts	Full enrichment, structured

flects the prevalence of such functions in standard-library and algorithmic-task settings rather than intentional curation; we treat it as an external-validity limitation (§5.5).

A static recursion-exclusion check at the curation stage rejects any function with self-recursion or mutually recursive helpers; the 90-day scope restricts the study to non-recursive code (Constitution §17.5).

For each function we prepare ground-truth artifacts: a single buggy variant whose defect falls into one of five categories (off-by-one, wrong comparison, missing boundary, incorrect variable, swapped arguments; Constitution §17.3), an annotation of the bug location and category, and a reference-fix test suite.

3.3 Information parity: operational definition and construction

We define the *information content* of a representation as the set of typed propositions derivable from it: the function signature types and the parameter and return contracts (precondition, postcondition, invariant). Under this definition C1+ and C4 are equivalent by construction.

C1+ is produced by projecting C4's annotations into Python source: PEP 484 inline type hints in the signature, plus a structured docstring block carrying the contract clauses in a fixed template. The projection is implemented in `src/pipeline/ir_to_annotated_text.py`.

We acknowledge a residual asymmetry. The projection does not preserve AST node-kind metadata, IR-level syntactic child-ordering, or explicit cross-reference structure among contract clauses, which become free-form docstring text. Under our operational definition these are *encoding*, not *information*; a more demanding definition—Shannon-style bit count, or recoverability of the source IR from the projected text—would assign C4 strictly higher information content. We do not adopt that definition, because the empirical question is whether structure aids reasoning *given the same* *propositional content*, not whether representations differ in arbitrary encoding bits. A round-trip validation measuring C4 $\rightarrow$ C1+ $\rightarrow$ C4$'$ structural recovery is identified as journal-version follow-on work (Appendix D).

Contracts for C4, and therefore for C1+, are generated by an LLM that is not used as a test subject and are reviewed by the author under the contract review protocol (Constitution §17.2): the allowed edits are factual correction, deletion of vacuously true clauses, and a single missing-clause addition. Provenance—raw, reviewed, and diff—is logged for every function.

3.4 Tasks and primary endpoint

Three task types are defined: T1 (code understanding), T2 (bug detection), and T3 (code transformation). T1 asks the model to summarize a function's behavior in natural language from its representation; T3 asks the model to produce a modified version of a function satisfying a contract-described change. Both are constitutional task families; the 90-day study collected only T2, so T1 and T3 are uncollected, and the six corresponding cells of the confirmatory family are insufficient-data placeholders (§3.6, §4.2).

T2 was pre-registered as the primary endpoint (Constitution §17.4) for objectivity: its 0–3 composite score—the sum of location, diagnosis, and fix sub-scores—is fully automated and requires no human judgment at scoring time. We honor that pre-registration throughout §4: H1 on T2 is the primary endpoint and carries the strongest interpretive weight.

3.5 Models and sampling regime

The confirmatory pair comprised `gpt-5.4` and `claude-opus-4-6`, sampled at `temperature` = 0. A post-confirmatory exploratory replication used the same-provider successor frontier pair—`gpt-5.5` and `claude-opus-4-7`—sampled at provider-default temperature. The successor models reject `temperature` = 0 at the API level; provider-default temperature is therefore a hard constraint, not a methodological preference.

This introduces a sampling-regime asymmetry: deterministic on the confirmatory pair, stochastic on the successor pair. We accordingly frame the cross-pair comparison as a robustness check under temperature relaxation, not as cross-generation replication; the two pairs are not directly comparable under a shared sampling regime. The successor pair was selected as the same-provider frontier successors so that representation effect, rather than provider drift, remains the axis of interest.

A brief disambiguation on terminology. The successor run is called a "replication" in the abstract and in §3.1 as a description of the *process*: a fresh run on a successor model pair, identified in the run logs as `t2_frontier_successor_replication`. What this subsection denies is "replication" as an *inferential framing* of the cross-pair evidence: the deterministic/stochastic asymmetry above precludes that framing, and §4.3 returns to the point. Both usages are correct; they refer to different things.

3.6 Statistical plan

The pre-registered confirmatory family comprises three hypotheses (H1: C4 > C1+; H2: C4 > C1; H3: C1+ > C1) crossed with three primary metrics (T1, T2, T3), giving nine tests. For each test we compute, per function, a paired difference between the two conditions after averaging the two model scores; we then apply the Wilcoxon signed-rank test with a one-sided "greater" alternative matching the pre-registered direction. The effect size is the rank-biserial correlation

$$r_{\text{rb}} \;=\; \frac{T^{+} - T^{-}}{T^{+} + T^{-}},$$

where T^+ and T^- are the sums of the positive and negative signed ranks; we use the abbreviation r_{rb} throughout §4 and §5. We apply the Holm–Bonferroni step-down correction across the nine-test family at $\alpha = 0.05$. Holm–Bonferroni is a family-wise-error-rate correction that orders the raw p-values from smallest to largest and tests each in turn against a tightening threshold (α/m at the smallest rank, $\alpha/(m-1)$ at the next, and so on up to α at the largest); a rank is rejected only if it and every earlier rank clear their thresholds. We also report a 95% percentile bootstrap CI on r_{rb} from 10,000 seeded resamples.

We honor $m = 9$ as pre-registered. Six of the nine cells are insufficient-data placeholders because T1 and T3 were not collected (§3.4); in reporting we distinguish the three collected-and-tested cells from the six scope-limited-uncollected cells, but we do not retroactively reduce m to 3. Post-hoc m-reduction would forfeit the value of pre-registration: the cost is reduced corrected power, the benefit is that the result speaks under the rules committed to before data were seen. Effective sample size per test is further compressed because Wilcoxon discards zero paired differences: against $n_{\text{pairs}} = 30$, the tie distribution induced by the rubric (§3.7) leaves an effective n of 13–14. The statistical plan also fixes the exploratory/confirmatory boundary so that downstream sections can quote it without re-deciding: per-model analyses are exploratory by Constitution §17.4 and are reported uncorrected with explicit exploratory labeling.

On bootstrap CI provenance: Constitution §17.4 mandates 95% bootstrap CIs on effect sizes. The frozen `analyze_confirmatory.py` does not compute CIs; CIs were computed via a sibling implementation verified against the frozen analyzer's point estimates. CI computation does not alter any hypothesis-testing decision; it adds reporting fidelity. Confirmatory CIs are reported in §4 alongside point estimates; provenance is recorded in Appendix E.

3.7 Score rubric and its empirical distribution

The T2 composite score is the integer sum of three binary sub-scores: location (0/1), diagnosis (0/1), and fix (0/1). Per-model support is therefore $\{0, 1, 2, 3\}$; the per-(function, condition) 2-model-averaged support is $\{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$.

We surface the empirical distribution here—before results—because it constrains what the analysis can detect. The confirmatory run contains 300 per-model T2 records (30 functions $\times$ 2 models $\times$ 5 conditions); their raw composite distribution is 0: 19 records (6%), 1: 89 (30%), 2: 192 (64%), and 3: 0 (0%). No model scored 3/3 on any T2 item; the location sub-score is the empirical bottleneck. After 2-model averaging, composite supports 2.5 and 3.0 are both empty, leaving an effective range of $[0.0, 2.0]$ with mode 2.0. Across H1, H2, and H3 at least 53% of paired differences are exactly zero; Wilcoxon discards these, and the effective n contracts to 13–14 from $n_{\text{pairs}} = 30$.

The ceiling is a property of the as-implemented rubric, not of the data. The location sub-score's saturation rate is a per-item property: each function independently lands somewhere in the $\{0, 1, 2, 3\}$ raw composite distribution shown above, and that distribution is determined by what the rubric can score, not by how many functions are sampled. Expanding n alone therefore does not solve underpowering—additional functions arrive drawn from the same distribution and produce zero-valued paired differences at the same rate, so the effective Wilcoxon n stays in the same fraction of n_{pairs}. The path forward (§6) is rubric refinement, or supplementing the composite with a continuous-valued T2 metric. Score histograms are shown in Figure 4.

3.8 Pre-registration

The Lumen Design Constitution v0.2.2—including the full statistical plan—was committed to the public repository before data collection began (commit `6b83bca`). The constitutional document at `docs/design-constitution.tex` was committed at repository scaffolding and has not been edited since, which strengthens the pre-registration claim against post-hoc adjustment. The remaining pre-registration timeline—analysis-machinery freeze, confirmatory closure, successor-plan registration, and exploratory tooling—is recorded with commit hashes in Appendix B.

4 Results

4.1 Confirmatory: H1/T2 primary endpoint

The primary endpoint of the study is the pre-registered H1 test on T2: under the C4-versus-C1+ comparison, does the structured-IR condition outperform its information-parity textual control on bug-detection composite score? Table 2 reports the full result.

Decision: fail to reject H_0 at $\alpha = 0.05$ corrected. With H2/T2 and H3/T2 also non-rejected at the family-corrected level (§4.2), the confirmatory family yields no rejection on T2. Under Constitution §9 the study therefore falls on the *Informative Negative Outcomes* path rather than meeting the Minimum-success threshold, which requires at least one of H1–H3 to be rejected at the corrected significance level. The materials are publicly released in accordance with the same criterion.

The non-rejection is informative rather than empty. The point estimate $r_{\text{rb}} = +0.581$ is moderate and lies in the pre-registered direction; the 95% percentile bootstrap CI $[+0.077, +0.917]$ excludes zero; and the one-sided raw $p = 0.023$ would reject H_0 in isolation. The failure occurs at the Holm–Bonferroni family-correction stage with $m = 9$: the corrected $p = 0.205$, and at the present effective Wilcoxon $n = 14$ (compressed from $n_{\text{pairs}} = 30$ by the rubric-induced tie distribution) the correction absorbs the moderate effect. The structural source of the under-rejection—the score-rubric ceiling—is documented in §3.7 and discussed in §5.3.

Table 2. H1/T2 primary-endpoint (*Confirmatory analysis*). Bootstrap parameters: seed 20260427, 10 000 resamples, percentile method. The 95% CI on r **excludes zero**.

Field	Value
Wilcoxon W	83.0
Raw p (one-sided "greater")	0.023
Holm-corrected p ($m = 9$)	0.205
Rank-biserial r	+0.581
95% bootstrap CI on r	**[+0.077, +0.917]**
n_{pairs}	30
n_+ / n_- / $n_=$	10 / 4 / 16
Effective Wilcoxon n	14
Median paired difference	0.0
Mean paired difference	+0.183
Decision: fail to reject H_0 at $\alpha = 0.05$ corrected.	

4.2 Confirmatory: secondary T2 layer and family completeness

Table 3 reports the full nine-test confirmatory family: three hypotheses crossed with three primary metrics, with $m = 9$ honored as pre-registered.

Of the three tested cells (all on T2), two yield rank-biserial r consistent with their pre-registered direction: H1/T2 with $r_{\text{rb}} = +0.581$ (CI $[+0.077, +0.917]$, raw $p = 0.023$) and H2/T2 with $r_{\text{rb}} = +0.200$ (CI $[-0.333, +0.700]$, raw $p = 0.233$). The third, H3/T2, yields $r_{\text{rb}} = -0.473$ (CI $[-0.885, +0.077]$, raw $p = 0.969$)—*opposite* the pre-registered direction. None of the three reaches Holm-corrected significance. The H3/T2 upper bound (+0.077) only just crosses zero.

The remaining six cells of the family are T1 and T3 hypotheses; T1 and T3 were uncollected within the 90-day scope (§3.4) and contribute Holm $p = 1.000$ by convention. We do not retroactively reduce m. The distinction between collected-and-tested and scope-limited-uncollected cells is a reporting matter (§3.6), not a multiplicity adjustment, and the six uncollected cells remain in the family.

The H3/T2 directional inversion is the strongest exploratory signal in the dataset; its per-model decomposition is taken up in §4.4.

Table 3. Confirmatory 9-test family: three tested T2 cells and six uncollected T1/T3 cells (*Confirmatory analysis*). $m = 9$ honored as pre-registered; uncollected cells contribute Holm $p = 1.000$ by convention.

Hyp	Task	n	W	raw p	Holm p ($m = 9$)	r_{rb}	95% CI	Status
H1	T1	—	—	—	1.000	—	—	uncollected
H1	T2	30	83.0	0.023	0.205	+0.581	$[+0.077, +0.917]$	tested
H1	T3	—	—	—	1.000	—	—	uncollected
H2	T1	—	—	—	1.000	—	—	uncollected
H2	T2	30	63.0	0.233	1.000	+0.200	$[-0.333, +0.700]$	tested
H2	T3	—	—	—	1.000	—	—	uncollected
H3	T1	—	—	—	1.000	—	—	uncollected
H3	T2	30	24.0	0.969	1.000	−0.473	$[-0.885, +0.077]$	tested
H3	T3	—	—	—	1.000	—	—	uncollected

4.3 Exploratory: H1 per-model direction concordance

The analyses in this and the following two subsections are exploratory by Constitution §17.4; they are reported uncorrected and are not elevated to confirmatory claims. The two per-model panels below report H1/T2 on the confirmatory and successor pairs in turn; their joint reading is then summarized.

4.3.1 Confirmatory pair, T2

Table 4 reports H1/T2 split by model on the confirmatory pair. `claude-opus-4-6` yields $r_{\mathrm{rb}} = +0.750$ with 95% CI $[+0.142, +1.000]$ excluding zero (raw one-sided $p = 0.055$); `gpt-5.4` yields $r_{\mathrm{rb}} = +0.394$ with CI $[-0.250, +0.864]$ (raw $p = 0.135$). Both per-model estimates point in the pre-registered direction.

Table 4. H1/T2 per-model rank-biserial, confirmatory pair (*Exploratory analysis*, uncorrected).

Model	n	r_{rb}	95% CI	raw p (1-sided)
`claude-opus-4-6`	30	**+0.750**	$[\mathbf{+0.142}, \mathbf{+1.000}]$	0.055
`gpt-5.4`	30	+0.394	$[-0.250, +0.864]$	0.135

4.3.2 Successor pair, T2 (under temperature relaxation)

Table 5 reports the same split on the successor pair, sampled at provider-default temperature (§3.5). `claude-opus-4-7` yields $r_{\mathrm{rb}} = +0.509$ (CI $[-0.143, +1.000]$, raw $p = 0.093$); `gpt-5.5` yields $r_{\mathrm{rb}} = +0.600$ (CI $[-0.333, +1.000]$, raw $p = 0.188$). The `gpt-5.5` row is computed on $n = 26$: twelve parse-failure exclusions in the successor run reduced its effective pair count, distributed across non-C1 conditions.

Across all four model-runs the H1 direction is consistent (per-model effect sizes and CIs in Tables 4 and 5). The successor pair preserves the direction under a relaxed sampling regime, so the H1 direction is not a single-model artifact and is not erased by stochastic sampling. We frame this as robustness evidence under temperature relaxation; the deterministic-versus-stochastic asymmetry between the two pairs (§3.5) precludes a cross-generation-replication framing. Figure 2 renders the four rows as a forest plot with a vertical zero line.

Table 5. H1/T2 per-model rank-biserial, successor pair (*Exploratory analysis*, uncorrected). Provider-default temperature. `gpt-5.5` $n = 26$ due to 12 parse-failure exclusions in the successor run, distributed across non-C1 conditions.

Model	n	r_{rb}	95% CI	raw p (1-sided)
`claude-opus-4-7`	30	$+0.509$	$[-0.143, +1.000]$	0.093
`gpt-5.5`	26	$+0.600$	$[-0.333, +1.000]$	0.188

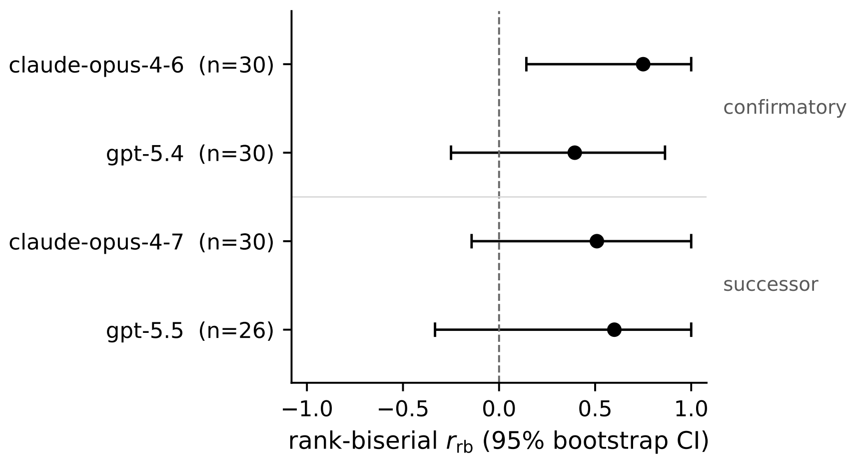

Figure 2. Forest plot of H1/T2 per-model rank-biserial with 95% CI across the confirmatory and successor pairs (4 rows). Vertical zero line indicated. *Exploratory analysis.*

4.4 Exploratory: H3 directional inversion across all four model-runs

All four model-runs show a negative H3 direction (with the caveat that the `claude-opus-4-7` row's r_{rb} is at the estimator's lower bound under a collapsed CI; see the in-line caveat below and the table footnote, and read the four-run count as direction sign only). The pre-registered prediction was $C1+ > C1$—that information enrichment in textual form would aid T2—but Table 6 shows C1+ underperforming C1 in every run: `claude-opus-4-6` $r_{\mathrm{rb}} = -0.333$, `gpt-5.4` -0.400, `claude-opus-4-7` -1.000, and `gpt-5.5` -0.500.

The `claude-opus-4-7` row warrants a caveat in line. Its $r_{\mathrm{rb}} = -1.000$ with collapsed 95% CI $[-1.000, -1.000]$ reflects 0 positive against 8 negative paired differences in 30 pairs (22 ties); under such sparse non-zero data the rank-biserial estimator saturates at its lower bound and the CI degenerates. This is an artifact of the estimator at small effective n, not a strong negative claim.

Treating each model-run's H3 sign as a single observation under a null of no effect gives $(1/2)^4 = 1/16 \approx 0.063$ for four negatives out of four. We report this as an upper-bound exploratory consistency check, not as an inferential test: the four runs share dataset and rubric, and the two members of each pair share inputs, so the runs are not strictly independent and the bound is loose.

An auxiliary observation from parse logs: `gpt-5.5` produced unparseable responses on 0/30 C1 items in the successor run but on 12/120 non-C1 items; among the non-C1 conditions it is the structured and enriched ones that destabilize the model's output. The instability is concentrated where the input departs from plain source.

This concentration also explains the per-pair n asymmetry between Table 5 and Table 6 on the `gpt-5.5` row. H1 compares C4 to C1+, both of which are non-C1 and so vulnerable to parse failure, and the row drops to $n = 26$; H3 compares C1+ to C1, and C1 always parses, so only C1+ failures reduce the pair count and the row stands at $n = 28$.

Taken together, the data exploratorily suggest a pattern about in-text annotation. On top of plain Python source, in-text type and contract annotations do not aid—and may impede—frontier LLM bug detection. The qualifier "at the current scoring rubric" is essential to the reading: the pattern is observed under the as-implemented T2 composite (§3.7), and a refined rubric could re-position the comparison. The pattern is consistent with, but does not confirm, the dissociation of representational structure from information content that the C1+ control was designed to test. Figure 3 renders the four rows as a forest plot and complements Figure 2.

Table 6. H3/T2 per-model rank-biserial across all four model-runs (*Exploratory analysis*). All four directions are negative. The `claude-opus-4-7` row has a collapsed CI at $[-1.000, -1.000]$ reflecting 0 positive vs 8 negative diffs in 30 pairs (22 ties); this is a CI artifact at small effective n, not a strong negative claim.

Run	Model	n_{pairs}	r_{rb}	95% CI	$n_+/n_-/n_=$
confirmatory v2	`claude-opus-4-6`	30	-0.333	$[-1.000, +0.333]$	3/6/21
confirmatory v2	`gpt-5.4`	30	-0.400	$[-1.000, +0.333]$	3/6/21
successor	`claude-opus-4-7`	30	-1.000	$[-1.000, -1.000]$	0/8/22
successor	`gpt-5.5`	28	-0.500	$[-1.000, +1.000]$	1/3/24

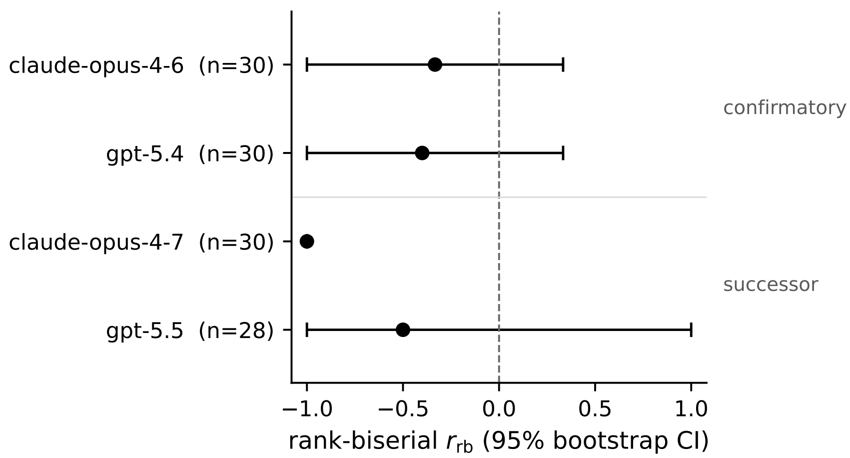

Figure 3. Forest plot of H3/T2 per-model rank-biserial with 95% CI across the confirmatory and successor pairs (4 rows). Vertical zero line indicated. Demonstrates the inverted-direction pattern visually; complements Figure 2. *Exploratory analysis.*

4.5 Exploratory: cumulative-decomposition E1–E3

Table 7 reports the cumulative-decomposition effect sizes for E1 (C2 vs C1), E2 (C3 vs C2), and E3 (C4 vs C3) on both runs; the full numerical table—with W, raw p, CI, and tie counts—is in Appendix C. On the confirmatory pair, r_{rb} values are E1 $+0.250$, E2 -0.200, E3 $+0.250$; on the successor pair, E1 -0.341, E2 $+0.250$, E3 -0.333.

No single decomposition step reaches significance in either run, and the direction inverts between the confirmatory and successor runs on E1 and E3. The cumulative chain $C1 \to C2 \to C3 \to C4$ does not exhibit a clean monotone build-up of effect through the four conditions.

This pattern, observed post hoc, is consistent with the pre-registered choice of central comparison: the systematic signal in the data is carried by the structure-versus-information axis (C4 vs C1+, isolated by construction) rather than by the marginal information additions along the cumulative chain.

Table 7. Cumulative-decomposition per-pair effect sizes (*Exploratory analysis*, 2-model averaged). Full numerical table in Appendix C.

Run	Hyp	Pair	r_{rb}	Direction
v2	E1	C2 vs C1	+0.250	+
v2	E2	C3 vs C2	−0.200	−
v2	E3	C4 vs C3	+0.250	+
successor	E1	C2 vs C1	−0.341	−
successor	E2	C3 vs C2	+0.250	+
successor	E3	C4 vs C3	−0.333	−

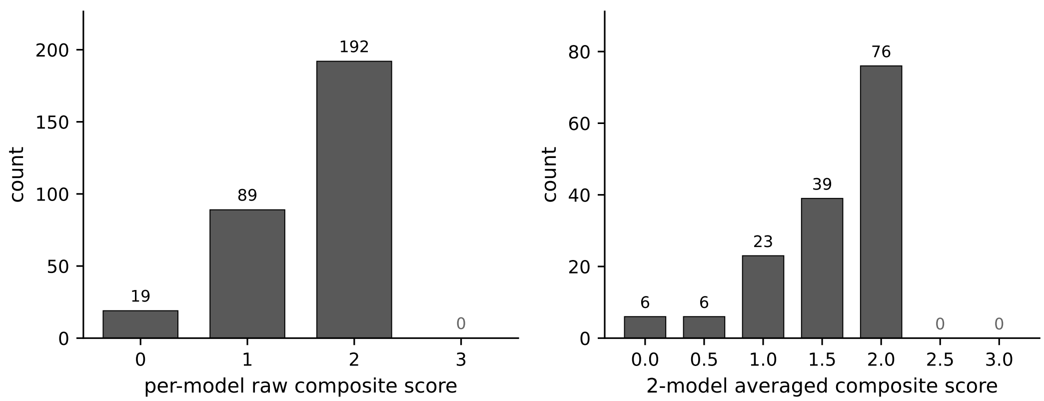

Figure 4. Score distribution histograms. Left: per-model raw composite $\{0, 1, 2, 3\}$ (composite 3 absent in 300/300 records). Right: per-(function, condition) 2-model-averaged composite $\{0, 0.5, \ldots, 3.0\}$ over all $30 \times 5 = 150$ cells (supports 2.5 and 3.0 both absent across all 150 cells). Visual support for §3.7.

5 Discussion

5.1 What the primary endpoint does and does not support

The non-rejection of H1/T2 at the Holm-corrected family level (§4.1) is not evidence of no effect. The honest reading of the primary endpoint splits across two clauses that must hold together; neither overrides the other.

First clause: the point estimate. The point estimate $r_{\text{rb}} = +0.581$ is moderate and lies in the pre-registered direction, and the one-sided raw $p = 0.023$ would reject H_0 in isolation. The data are consistent with the pre-registered direction at moderate effect size.

Second clause: the family correction. The failure occurs at the Holm–Bonferroni family-correction stage with $m = 9$ (§3.6), six of whose cells are insufficient-data placeholders contributing Holm $p = 1.000$ by convention; the corrected $p = 0.205$; and the effective Wilcoxon $n \approx 14$, compressed from $n_{\text{pairs}} = 30$ by the rubric-induced tie distribution, does not support a confirmatory claim.

The CI excluding zero qualifies the non-rejection; it does not overturn it.

A back-of-the-envelope point reinforces the second clause. At the most-promising rank in $m = 9$, the Holm-corrected α is $\alpha/m = 0.05/9 \approx 0.0056$; detecting $r \approx +0.58$ at that threshold would require substantially larger non-tied effective n than the present data provide. The implication, taken up in §5.3 and §6.2, is that rubric refinement, not n alone, is the path forward.

5.2 Cross-run direction concordance: what it permits and forbids

The H1 direction is positive in all four model-runs across the confirmatory and successor pairs (§4.3); the H3 direction is negative in all four (§4.4). These analyses are exploratory under Constitution §17.4, uncorrected, and underpowered per cell, and cannot be promoted to confirmatory claims.

What the concordance *permits.* It constrains the explanation space. The positive H1 direction is not a single-model artifact, and it survives a relaxed sampling regime on the successor pair. The H3 inversion is not noise on one run; it appears in every run. Together these patterns shift the prior on what a properly powered confirmatory study would find, in opposite directions for the two hypotheses. What the concordance *forbids.* It does not promote H1 to confirmatory status, and it does not establish the H3 inversion as a genuine direction. Multiplicity is unaddressed; the four model-runs share dataset, contracts, and rubric, so cross-run agreement does not deliver the independence a confirmatory test would require. The exploratory signals are inputs to follow-on design (§6.2), not substitutes for it.

5.3 The score-rubric ceiling is a structural finding, not a side issue

The location sub-score's empirical near-saturation (§3.7) is the binding constraint on confirmatory power in this study. The rubric assigns 0, 1, or 2 points to the location component of a T2 verdict; the saturation rate at the 2-point ceiling, documented in the empirical distribution of §3.7, leaves almost no headroom for C4 to outscore C1+ on this component within a paired comparison. The compressed effective Wilcoxon n on H1/T2—14 against $n_{\text{pairs}} = 30$ (§4.1)—is the direct consequence: 16 of 30 paired differences tie at zero, and the moderate point estimate must clear Holm correction with that compressed denominator.

This is a property of the as-implemented T2 rubric, not of the dataset or the models, and the distinction determines the path forward. Scaling n from 30 to 50 or 100 would not change the saturation rate; it would multiply tied paired differences in the same proportion, leave effective n in the same fraction, and leave the Holm-corrected α-budget no easier to meet. n expansion is necessary, but it is not sufficient and not first. The follow-on (§6.2) must refine the location sub-score—finer ordinal categories, or a continuous-valued T2 metric supplementing the composite—before expanding n. We state this as a methodological finding with reuse value beyond the present study: in any composite-rubric, paired-comparison design where one sub-score saturates against its ceiling, n expansion alone will be a costly and ineffective response to underpowering.

5.4 Implications for the long-term Lumen vision

Constitution v0.2.2 §3 specifies a reassessment threshold for the canonical-core-over-surface-text bet—the design wager that a structured canonical core IR with text and LLM-facing notations as projections will outperform surface-text representations for code reasoning. The threshold: an outcome of $C1+ \geq C4$ on all primary metrics across both confirmatory models would trigger formal reassessment. The observed outcome on H1/T2—$C4 > C1+$ in direction, moderate effect size, failing Holm correction at $m = 9$—does not meet that threshold and therefore does not trigger reassessment.

It also does not constitute confirmatory support for the bet. The point estimate is consistent with the bet's direction; the statistical confirmation it would require has not been delivered. The empirical motivation for the bet is intact at the level of point estimates and weakened at the level of statistical confirmation. The project continues at the architectural level pending T1 and T3 collection, rubric refinement (§5.3), and a properly powered confirmatory re-test (§6.2). Neither overclaim nor write-off is warranted; both would misrepresent the data.

5.5 Threats to validity (consolidated)

Several threats remain on the table and bound what the present results support.

- **Power.** $n_{\text{pairs}} = 30$ with effective Wilcoxon $n = 13$–14 after rubric-induced ties (§3.6, §3.7) bounds power on every confirmatory cell.
- **Phase-2 power analysis not produced.** Constitution §16 (Open Question 4) called for a Phase-2 pilot power analysis or effect-size sanity check before fixing n_{pairs}; this artifact was not produced (§3.2, Appendix E), and the rubric-induced compression of effective n was therefore identified post hoc rather than pre-confirmatory.
- **Family completeness.** T1 and T3 were uncollected within the 90-day scope; six of the nine confirmatory family cells are scope-limited (§4.2), and the results do not generalize to other task families.
- **Dataset domain skew.** Approximately two-thirds (20 of 30) of the retained functions are list and array operations; results may not generalize uniformly to arithmetic, parsing, or string-heavy domains.
- **Author-reviewed contracts.** Single-evaluator residual variance enters the rubric inputs. For H1 (C4 vs C1+) the same reviewed contracts are present in both conditions, so reviewer-side variance cancels in the paired comparison; for H3 (C1+ vs C1) the contracts are present on C1+ only, and any systematic reviewer bias does not cancel across the pair.
- **Two providers, two generations.** The frontier-LLM landscape is not fully covered.
- **Non-recursive single-function scope.** A Constitution-mandated 90-day scope restriction.
- **Information parity is operational, not absolute.** The residual asymmetry note in §3.3 stands; round-trip C4 $\rightarrow$ C1+ $\rightarrow$ C4$'$ validation is journal-version follow-on work.

6 Conclusion

6.1 Summary

We proposed a methodology for information-parity testing of code-representation effects on frozen frontier LLMs and instantiated it on T2 bug detection. On the pre-registered primary endpoint the result is *direction-consistent but Holm-non-significant*: the point estimate $r_{\text{rb}} = +0.581$ lies in the pre-registered direction, but the corrected $p = 0.205$ at $m = 9$ does not reject H_0. Exploratory analyses find the H1 sign positive in all four model-runs (including the successor pair under a relaxed sampling regime; §3.6) and a systematic H3 sign inversion across the same four model-runs—reported under exploratory labeling, not promoted to confirmatory claims. Post-hoc inspection identifies a scoring-rubric ceiling as the structural source of confirmatory underpowering.

6.2 Immediate follow-on

Several concrete steps follow from the present results. They are listed in priority order, with the first item as the binding constraint to address before any dataset expansion.

- **Refine the T2 scoring rubric** to address the location-sub-score saturation (§3.7, §5.3). This is the binding constraint on confirmatory power and precedes n expansion.
- **Collect T1 and T3 data** under a separate confirmatory pre-registration, so that the six insufficient-data cells of the nine-test family (§4.2) become testable.

- **Implement the C4 $\rightarrow$ C1+ $\rightarrow$ C4′ round-trip validation** (§3.3, Appendix D) to quantify the information-parity residuals beyond the operational definition adopted here.
- **Back-port the bootstrap CI** into the frozen `analyze_confirmatory.py` via a structurally equivalent re-implementation, now that the sibling CI implementation has been verified against the frozen point estimates (§3.6, Appendix E).
- **Expand** n (e.g. to 50) under separate pre-registration—*after* rubric refinement, not before (§5.3).
- **Broaden dataset domain coverage** to dilute the list-operations skew documented in §5.5.

6.3 Long-term

The study tests a precondition for the broader Lumen vision — a canonical core IR with text and LLM-facing notations as projections: namely, that representation form materially affects external-agent reasoning. The direction-consistent confirmatory point estimate, together with the exploratory agreement of the H1 sign across all four model-runs (the successor pair under the relaxed sampling regime of §3.6, not a cross-generation replication), keeps that precondition *plausible*; it does not establish it. A definitive empirical case awaits T1 and T3 collection and rubric refinement; the canonical-core-over-surface-text bet (§5.4) remains active, and is reassessed only on stronger contrary evidence than the present data provide.

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Marjane Namavar, Noor Nashid, and Ali Mesbah. A controlled experiment of different code representations for learning-based program repair. *Empirical Software Engineering*, 27(7), 2022. doi: 10.1007/s10664-022-10223-5.

Abel Salinas and Fred Morstatter. The butterfly effect of altering prompts: How small changes and jailbreaks affect large language model performance. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024.

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Appendices

A Retained 30 `func_ids` and domain categorization

The 30 retained non-recursive Python functions used for the confirmatory and successor T2 runs are listed below in the order recorded by `input_summary.func_ids_used`.

Table 8. Retained 30 `func_ids` with their domain categorization in {`arithmetic`, `list_operations`, `string_processing`, `statistics`, `control_flow`}. Categories derived from each function's signature and docstring in `data/functions/raw/`.

#	func_id	Domain
1	antidiagonals	list_operations
2	batch_list	list_operations
3	camel_to_snake	string_processing
4	chunk_on_change	list_operations
5	count_true_segments	list_operations
6	dense_rank	statistics
7	equal_width_buckets	statistics
8	find_balanced_spans	control_flow
9	first_index_of_max	list_operations
10	frequency_table	statistics
11	group_and_aggregate	statistics
12	group_by_key	list_operations
13	longest_plateau	list_operations
14	max_subarray_bounds	arithmetic
15	merge_intervals	list_operations
16	merge_sorted	list_operations
17	peak_valley_indices	list_operations
18	remove_adjacent_dups	list_operations
19	rle_encode	list_operations
20	rotate_list	list_operations
21	segments_above_threshold	list_operations
22	sliding_window_max	list_operations
23	sorted_list_intersection	list_operations
24	spiral_order	list_operations
25	stable_partition	list_operations
26	strided_windows	list_operations
27	tokenize_arithmetic	string_processing
28	top_k_by	statistics
29	two_sum_sorted_pairs	list_operations
30	welford_running_stats	statistics

Aggregate. `arithmetic` 1, `list_operations` 20, `string_processing` 2, `statistics` 6, `control_flow` 1; total 30.

Curation rule. A static recursion-exclusion check at the dataset curation stage rejects any function with self-recursion or mutually recursive helpers; the 90-day study scope explicitly restricts to non-recursive code (Constitution §17.5).

B Pre-registration timeline and key commit hashes

```
Pre-registration                  : Lumen Design Constitution v0.2.2
```

```
                                commit 6b83bca
                                (docs/design-constitution.tex was
                                committed at repository scaffolding
                                and never edited thereafter; this
                                is the sole commit touching that
                                file in the git log)

Analysis machinery freeze     : 9217a26
                                phase7b: freeze canonical execution
                                state for full T2 confirmatory v2 rerun

Confirmatory closure          : 10beff2
                                phase7c: close confirmatory T2
                                full_t2_confirmatory_v2 authoritative

Successor plan registration   : 9c91347
                                phase7d: register T2 frontier
                                successor replication plan

Exploratory tooling           : 865fbe0
                                phase7e: close T2 frontier successor
                                exploratory replication

Phase 7e priority 2           : 54e722c
                                phase7e-p2: implement
                                analyze_exploratory.py + tests
```

C Full E1–E3 results, confirmatory v2 and successor runs

Table 9. Full E1–E3 cumulative-decomposition results, confirmatory v2 run (`full_t2_confirmatory_v2`, 2-model averaged) and successor run (`t2_frontier_successor_replication`, 2-model averaged). *Exploratory analysis.* Each row pairs the cumulative step (C2 vs C1, C3 vs C2, C4 vs C3) on T2.

Run	Step	n_{pairs}	W	raw p	r_{rb}	95% CI	$n_+/n_-/n_=$
v2	E1 (C2 vs C1)	30	75.0	0.173	+0.250	$[-0.273, +0.732]$	9/6/15
v2	E2 (C3 vs C2)	30	22.0	0.828	−0.200	$[-0.778, +0.429]$	4/6/20
v2	E3 (C4 vs C3)	30	22.5	0.363	+0.250	$[-0.500, +1.000]$	5/3/22
successor	E1 (C2 vs C1)	30	30.0	0.915	−0.341	$[-0.806, +0.238]$	5/8/17
successor	E2 (C3 vs C2)	30	17.5	0.383	+0.250	$[-0.667, +1.000]$	4/3/23
successor	E3 (C4 vs C3)	30	5.0	0.844	−0.333	$[-1.000, +1.000]$	2/3/25

Notes. Raw p-values are one-sided "greater" matching the pre-registered direction of each cumulative step. W is the Wilcoxon signed-rank statistic. Confidence intervals are 95% percentile bootstrap (seed 20260427, 10 000 resamples). The high tie counts ($n_= \in \{15, 17, 20, 22, 23, 25\}$) are the same effective-n compression phenomenon documented for the confirmatory family in §3.7.

D Information-parity round-trip validation

This appendix will report the C4 → C1+ → C4′ round-trip validation experiment in the journal version. Within the workshop scope, see §3.3 for the operational definition of information parity used by this paper.

E Reproducibility manifest

```
Repository
    git@github.com:WhiteStoneTak/Lumen.git
    HEAD at submission preparation: fca5e85

Key commit hashes (also Appendix B)
    Pre-registration             : Constitution v0.2.2
                                   (committed before data collection)
    Analysis machinery freeze    : 9217a26
    Confirmatory closure         : 10beff2
    Successor plan registration  : 9c91347
    Exploratory tooling          : 865fbe0
    analyze_exploratory.py       : 54e722c

Run wall-clocks
    full_t2_confirmatory_v2           : 36 min 13 s
    t2_frontier_successor_replication : 45 min 25 s

API endpoint snapshot dates
    Data collection day (both runs)   : 2026-04-27 (local time)
        Confirmatory : 2026-04-27 00:44 -- 01:21 (36 min 13 s)
        Successor    : 2026-04-27 09:09 -- 09:55 (45 min 25 s)
    Requested model_ids (logical, as passed to the provider SDKs)
        Confirmatory : claude-opus-4-6, gpt-5.4
        Successor    : claude-opus-4-7, gpt-5.5
    Provider-side dated snapshot identifiers (e.g., Anthropic dated
    suffix, OpenAI system_fingerprint) were not captured at
    collection time and are not recoverable from the saved response
    archives, which store only model_id, response_text, prompt_tokens,
    total_tokens, and error. We declare this as a provenance
    limitation; future runs should record these fields at the SDK
    call site.

Raw response archives
    Confirmatory : results/runs/full_t2_confirmatory_v2/responses/
    Successor    : results/runs/t2_frontier_successor_replication/responses/
    (Score JSONs alongside, under .../scores/.)

Pre-confirmatory dataset-sufficiency audit (referenced by ğ3.1, ğ5.5)
    Path        : results/analysis/t2_dataset_sufficiency_audit_2026-04-02.json
    Schema      : t2-dataset-sufficiency-audit-v1
    Audit date  : 2026-04-02 (three weeks before full_t2_confirmatory_v2
```

```
                collection on 2026-04-27)
    Key finding : dataset_sufficiency_finding records
                  "minimum_line_reached": true,
                  "sufficiency_proven": false, and
                  "evidence_for_30_vs_50":
                    "No in-repo pilot power analysis artifact or
                     effect-size sanity-check artifact determining
                     30 vs 50 sufficiency was found."
    Decision    : branch_decision records that confirmatory execution
                  proceeded at the constitutional minimum line of
                  n = 30 (wave2 pool exhausted), with rubric-induced
                  effective-n compression identified post hoc from
                  confirmatory data (ğ3.7, ğ5.3).

Bootstrap CI scratch-script provenance (confirmatory effect sizes)
    Script        : scratch/bootstrap_ci_confirmatory.py
    Results JSON  : scratch/bootstrap_ci_results.json
    RNG seed      : 20260427
    Resamples     : 10,000
    Method        : percentile bootstrap on rank-biserial r
    Verification  : point estimates re-computed by the script and
                    matched analyze_confirmatory.py output to <0.001
                    on r and to identical W and raw p-values on the
                    Wilcoxon decisions.

    The frozen analyze_confirmatory.py does not compute CI; the scratch
    script is a sibling implementation that adds CI reporting only and
    does not modify any pre-registered hypothesis-testing decision.

Figure provenance
    F1 (rendered as Figure 2; H1/T2 per-model forest)
        Script   : paper/figures/scripts/make_f1_f2_forest.py
        Source   : results/analysis/exploratory/
                       full_t2_confirmatory_v2/exploratory_results.json
                   results/analysis/exploratory/
                       t2_frontier_successor_replication/exploratory_results.json
                   (per_model_split_h1_h2_h3; H1, T2 entries)
        Output   : paper/figures/F1.pdf
        Recompute: python3 paper/figures/scripts/make_f1_f2_forest.py
        No statistics are recomputed; r and 95% bootstrap CI are read
        verbatim from the analysis artifacts and plotted as-is.
        Script asserts each row's r and CI against the values in
        Tables 4-5 of the paper.

    F2 (rendered as Figure 3; H3/T2 per-model forest)
        Same script and same source files as F1
        (per_model_split_h1_h2_h3; H3, T2 entries).
        Output   : paper/figures/F2.pdf
```

```
    Script asserts each row against Table 6, including the collapsed
    CI [-1.000, -1.000] on the claude-opus-4-7 row.

F3 (rendered as Figure 4; T2 score distribution histograms)
    Script   : paper/figures/scripts/make_f3_histograms.py
    Source   : results/runs/full_t2_confirmatory_v2/scores/*.json
               (scorer-result-v1; 300 records)
    Output   : paper/figures/F3.pdf
    Descriptive count over the frozen confirmatory run (no
    hypothesis test). Left panel = per-model raw composite over
    300 records; right panel = per-(function, condition) 2-model
    averaged composite over the 30 x 5 = 150 cells.
    Script asserts the left-panel counts against section 3.7
    ({0:19, 1:89, 2:192, 3:0}; total 300) and that the right-panel
    2.5 and 3.0 supports are empty.

F4 (rendered as Figure 1; 5-condition schematic)
    No data source; drawn inline in paper/main.tex as TikZ.
```